Exno : 08
Rollno: 241001051

Date: 25/09/2026

Backward Chaining

Exercise 1: Medical Diagnosis

```
% Facts
has_fever(john).
has_cough(john).
% Rule
has_flu(X) :- has_fever(X), has_cough(X).
% Query
% ?- has_flu(john).
```

Output:

true.

Exercise 2: Student Eligibility Checker

```
% Facts
passed(alice, maths).
passed(alice, science).
% Rule
eligible(X) :- passed(X, maths), passed(X, science).
% Query
% ?- eligible(alice).
```

Output:

true.

Exercise 3: Travel Planner

```
% Facts
```

has_ticket(james).

has_visa(james).

% Rule

can_travel(X) :- has_ticket(X), has_visa(X).

% Query

% ?- can_travel(james).

Output:

true.

Result:

Thus the program has been executed and the output is verified successfully.

```
File  Settings  Tools  Debug  GUI  Help
Welcome to SWI-Prolog (threaded, 64 bits, version 10.1.13)
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For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

18 ?- consult(['C:/Users/Lenovo/Documents/expno 8.pl']).
true.

19 ?- consult(['C:/Users/Lenovo/Documents/expno 9.pl']).
true.

20 ?- consult(['C:/Users/Lenovo/Documents/expno 10.pl']).
true.

21 ?- can_travel(james).
true.

22 ?- eligible(alice).
true.

23 ?- has_flu(john).
true.

24 ?-
```